

inference Engineering Assignment

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1 question 1

Why is inference memory bandwidth-bound important?

inference memory is bandwidth bound because the bandwidth decides the rate at which the information will be passed down from the memory to the GPU'S computing unit, The faster the information is passed down the faster the tokens will be generated by the LLM. if the LLM is able to compute faster but the speed at which it is recieveing information is slower, it will create a bottleneck. higher the memory bandwidth, better is the inference speed and throughput.

2 Question 2

How can we calculate the overall memory required to store a model?

How are LLM models represented while storing?

The overall memory required for an LLM during inference mainly depends on the model weights, KV cache, and additional runtime memory.

The approximate memory requirement can be represented as:

$$TotalMemory \approx WeightMemory + KVCache + RuntimeMemory$$

The memory required to store the model weights can be calculated as:

$$WeightMemory = NumberofParameters \times BytesperParameter$$

LLM models are stored using numerical data types such as FP32, FP16, BF16, INT8, and INT4. Lower-precision formats require less memory and can improve inference efficiency, although they may result in some loss of numerical precision.

3 Question 3

Explain each layer of inferencing and its importance to an institution/company.

in inference engineering thee are three layers to inferencing, Runtime,Infrastructure,Tooling

3.1 Runtime

Runtime is the software layer that actually executes the model on the available hardware. It handles model execution, memory management, and the generation of tokens.

An efficient runtime can reduce latency, improve throughput, and make better use of available GPU resources. This can reduce the cost of serving an LLM

3.2 Infrastructure

infrastructure layer consists of the hardware required. it usually consists of GPUs, VRAM, CPUs, RAM, storage, and networking. infrastructure is very important for the company as a good infrastructure decides how the model is able to keep up with the incoming requests, and handling all the load at the same time. The infrastructure should not collapse if the number of requests have increased, it should manage the requests and function normally which will take place with the help of tooling layer

3.3 Tooling

Tooling layer consists of the software used for optimizing the LLM and managing inference, it consists of the following softwares: TensorRT-LLM, vLLM, and SGLang. vllm is used for managing the incoming load, it uses the techniques such as PagedAttention and Continuous batching to improve the efficiency of the llm in delivering the response. SGLang acts as an inference runtime that manages the execution of LLM requests and optimizes GPU and memory utilization. TensorRT-LLM was created by NVIDIA to optimize the GPU at its peak for better performance

These layers are non-replaceable, in order for a company to run smoothly, these layers must be applied. A good LLM when deployed works only because of these layers. these layers are dependent on each other. If even one layer is not working properly, all the other layers will get affected too. For a company to make a Good profit from their model they need all 3 layers of inference.

4 Question 4

Pick any 2 GPUs. Find their VRAM, bandwidth, and rental cost. Calculate the biggest model each can hold at 16-bit and 4-bit.

4.1 NVIDIA RTX 5090

The NVIDIA RTX 5090 is a high-end graphics card from NVIDIA that is designed for gaming, artificial intelligence, machine learning, and other demanding

applications. It comes with 32 GB of GDDR7 VRAM and has a memory bandwidth of 1792 GB/s. The large memory capacity makes it suitable for handling AI models and other memory-intensive workloads.

For AI inference, the RTX 5090 can theoretically support a model of approximately 16 billion parameters at 16-bit precision. When using 4-bit precision, it can theoretically support models of approximately 64 billion parameters. These values are based on the available VRAM and are theoretical estimates.

4.2 NVIDIA RTX 3090

The NVIDIA RTX 3090 is a high-performance graphics card from NVIDIA that can be used for gaming, artificial intelligence, machine learning, and other demanding tasks. It comes with 24 GB of GDDR6X VRAM and has a memory bandwidth of 936 GB/s. Its large amount of VRAM makes it suitable for running AI models and performing inference tasks.

For AI inference, the RTX 3090 can theoretically support a model of approximately 12 billion parameters at 16-bit precision. At 4-bit precision, it can theoretically support models of approximately 48 billion parameters. These values are theoretical estimates based on the available VRAM.

5 Question 5

Research 2 serving frameworks we didn't cover in depth (TGI, Ollama, LMDeploy, etc.). For each, write a short summary on what it's good for and when you'd pick it over vLLM/SGLang/TensorRT-LLM.

6 Serving Frameworks

6.1 LMDeploy

LMDeploy is a serving framework designed for deploying and running large language models efficiently. It provides tools for model inference and supports features such as quantization and high-speed model serving.

LMDeploy is a good choice when efficient model deployment and quantization are important. It can be useful for applications that need to run language models with limited computing resources.

6.2 OpenLLM

OpenLLM is a framework that helps developers deploy and run large language models in applications. It provides a simple way to make models available through APIs and can be used with different types of language models.

OpenLLM is a good choice when developers want a straightforward way to deploy models and integrate them into their applications. It can be useful for experimenting with different models and building AI-based applications.

7 Question 6

Run `nvidia-smi` on any GPU machine and explain at least 5 things in the output.

8 NVIDIA-SMI Command

The `nvidia-smi` command was run on a GPU machine with an NVIDIA Tesla T4 GPU. It displays important information about the GPU and its current usage.

The following are some of the details shown in the output:

1. **GPU Name:** The GPU detected by the system is an NVIDIA Tesla T4.
2. **Driver Version:** The installed NVIDIA driver version is 580.82.07. The driver allows the operating system and applications to communicate with the GPU.
3. **CUDA Version:** The output shows CUDA Version 13.0, which indicates the CUDA version supported by the installed NVIDIA driver.
4. **Temperature:** The GPU temperature is 36°C at the time the command was executed.
5. **Memory Usage:** The GPU has a total of 15360 MiB of memory, out of which only 3 MiB was being used when the command was executed.
6. **GPU Utilization:** The GPU utilization is shown as 0%, meaning that the GPU was not performing any significant processing task at that time.
7. **Power Usage:** The GPU was using 9W of power out of a maximum capacity of 70W.
8. **Running Processes:** The output shows that there were no running processes using the GPU at the time of execution.

Overall, `nvidia-smi` provides a quick way to check the GPU's specifications, current memory usage, temperature, power usage, and active processes.